

Practical 12(R-K Method 4thOrder)

■ Question1

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In[ ]:= RK4thOrder[a0_, b0_, n0_, f_, alpha_, actualSolution_] :=  
Module[{a = a0, b = b0, n = n0, h, ti, K1, K2, K3, K4},  
  h = (b - a) / n;  
  ti = Table[a + (j - 1) h, {j, 1, n + 1}];  
  wi = Table[0, {n + 1}]; wi[[1]] = alpha;  
  actualSol = actualSolution[ti[[1]]];  
  difference = Abs[actualSol - wi[[1]]];  
  OutputDetails = {{0, ti[[1]], alpha, actualSol, difference}};  
  For[i = 1, i ≤ n, i++,  
    K1 = h * f[ti[[i]], wi[[i]]];  
    K2 = h * f[ti[[i]] + h / 2, wi[[i]] + K1 / 2];  
    K3 = h * f[ti[[i]] + h / 2, wi[[i]] + K2 / 2];  
    K4 = h * f[ti[[i]] + h, wi[[i]] + K3];  
  
    wi[[i + 1]] = wi[[i]] + (1 / 6) * (K1 + 2 K2 + 2 K3 + K4);  
  
    actualSol = actualSolution[ti[[i + 1]]];  
    difference = Abs[actualSol - wi[[i + 1]]];  
    OutputDetails = Append[OutputDetails,  
      {i, N[ti[[i + 1]]], N[wi[[i + 1]]], N[actualSol], N[difference]}}];  
  Print[NumberForm[TableForm[OutputDetails, TableHeadings →  
    {None, {"i", "ti", "wi", "actSol(ti)", "Abs(wi-actSol(ti))"}}, 6]]];  
f[t_, x_] := 1 + x / t;  
actualSolution[t_] := t (1 + Log[t]);  
RK4thOrder[1, 6, 10, f, 1, actualSolution]
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i	ti	wi	actSol(ti)	Abs(wi-actSol(ti))
0	1	1	1	0
1	1.5	2.10778	2.1082	0.000419884
2	2.	3.38563	3.38629	0.000668889
3	2.5	4.78985	4.79073	0.000876059
4	3.	6.29477	6.29584	0.0010692
5	3.5	7.88341	7.88467	0.00125661
6	4.	9.54374	9.54518	0.00144132
7	4.5	11.2667	11.2683	0.00162464
8	5.	13.0454	13.0472	0.00180718
9	5.5	14.8741	14.8761	0.00198926
10	6.	16.7484	16.7506	0.00217104